APPLE ITUNES MUSIC ANALYSIS

(SQL-Based Analytical Pipeline & Business Intelligence Dasboard)

Internship Project Report Submitted To
Innovexis – Apple iTunes Music Analysis Project

Submitted By:

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TABLE OF CONTENTS

SR NO	SECTION NAME	PAGE NO
1.	PROJECT OEVRVIEW	2
2.	PROBLEM STATEMENT	2
3.	EXECUTIVE SUMMARY	3
4.	BUSINESS GOALS	4
5.	TOOLS & TECHNOLOGIES	4
6.	DATABASE SETUP	5
7.	DATA IMPORT PROCESS USING IMPORT WIZARD	8
8.	DATA VALIDATION (Database Testing)	12
9.	EXPLORATORY DATA ANALYSIS(SQL QUERIES)	15
10.	ADVANCED SQL ANALYTICS	20
11.	ADVANCED ANALYTICS INSIGHTS	24
12.	BUSINESS INSIGHTS & RECOMMENDATIONS	25
13.	DASHBOARD CREATION (Power BI)	26
14	FINAL INSIGHTS & STRATEGIC RECOMMENDATIONS	37
15.	CONCLUSION	40

Project Title: Apple iTunes Music Analysis

REPORT

(SQL-Based Analytical Pipeline & Business Intelligence Dashboard)

Bhuimka Bhai Patil

1. PROJECT OEVRVIEW

The **Apple iTunes Music Analysis** project focuses on analyzing the operational and sales data of a digital music store to generate meaningful business insights. The iTunes platform manages a vast collection of music tracks, albums, artists, genres, and customers across multiple countries. As the business grows, understanding customer behavior, revenue trends, and product performance becomes essential for strategic decision-making.

In this project, a relational database was designed and implemented using structured CSV datasets containing information about customers, employees, invoices, tracks, albums, artists, genres, playlists, and media types. SQL was used to perform exploratory and advanced analytical queries to uncover patterns in purchasing behavior, revenue generation, and content popularity.

Additionally, an interactive business dashboard was created using **Microsoft Power BI Desktop** to visualize key performance indicators (KPIs) such as:

- Total Revenue
- Monthly Revenue Trends
- Top Customers
- Most Purchased Genres

The objective of this project is to transform raw transactional data into actionable business insights that can support marketing strategies, customer engagement initiatives, and product optimization decisions.

This end-to-end analytical pipeline demonstrates practical skills in database design, SQL analytics, data visualization, and business reporting.

2. PROBLEM STATEMENT

Apple iTunes maintains a large digital music store with millions of tracks, thousands of customers worldwide, and a network of employees managing sales operations. As the business expands, the leadership team is looking to gain deeper insights into customer behavior, music preferences, and overall sales performance.

As a Data Analyst, you are tasked with analyzing the iTunes relational database (provided in CSV format) to generate actionable insights that can help improve product offerings, customer targeting, and operational efficiency.

Your role is to build a complete SQL-based analytical pipeline using the available datasets, which include details about customers, employees, invoices, tracks, albums, artists, genres, playlists, and media types.

3.EXECUTIVE SUMMARY

The **Apple iTunes Music Analysis** project was undertaken to analyze transactional and customer data from a digital music store database and generate actionable business insights. Using structured CSV datasets, a relational database schema was designed and implemented with proper primary and foreign key relationships to ensure data integrity and consistency.

SQL was used to perform exploratory and advanced analysis, including revenue tracking, customer segmentation, genre popularity analysis, and regional sales evaluation. Techniques such as aggregation, joins, subqueries, and ranking functions were applied to extract meaningful patterns from the data.

Key findings revealed that revenue is concentrated among a small group of high-value customers and specific music genres. Monthly revenue trends indicated seasonal fluctuations, while regional analysis showed that certain countries contribute significantly more to overall sales. Additionally, a limited number of artists and genres drive the majority of purchases, highlighting opportunities for targeted promotions and catalog optimization.

An interactive dashboard was developed using **Microsoft Power BI Desktop** to visualize key performance indicators (KPIs) such as Total Revenue, Total Orders, Total Customers, Monthly Revenue Trends, Top Customers, and Most Purchased Genres. The dashboard enables quick decision-making by presenting complex analytical results in a user-friendly format.

Based on the analysis, strategic recommendations were proposed for marketing, product, and operations teams. These include implementing customer loyalty programs, promoting high-performing genres, optimizing underperforming content, and expanding targeted regional marketing efforts.

Overall, this project demonstrates the effective integration of database design, SQL analytics, and data visualization to support data-driven business decision-making. The end-to-end analytical pipeline highlights practical skills essential for real-world data analytics roles.

4. BUSINESS GOALS

1. Understand customer behavior and purchasing trends.
2. Identify the most and least popular music genres, tracks, and artists.
3. Evaluate sales performance by employees and customer regions.
4. Analyze revenue trends across time and product types (media types).

5. Uncover growth opportunities by identifying underutilized content or inactive customers.

5. TOOLS & TECHNOLOGIES

- SQL (PostgreSQL / MySQL / SQLite)
- CSV files for dataset import
- Optional: Tableau, Power BI for visualization

- GitHub or Google Docs for project documentation

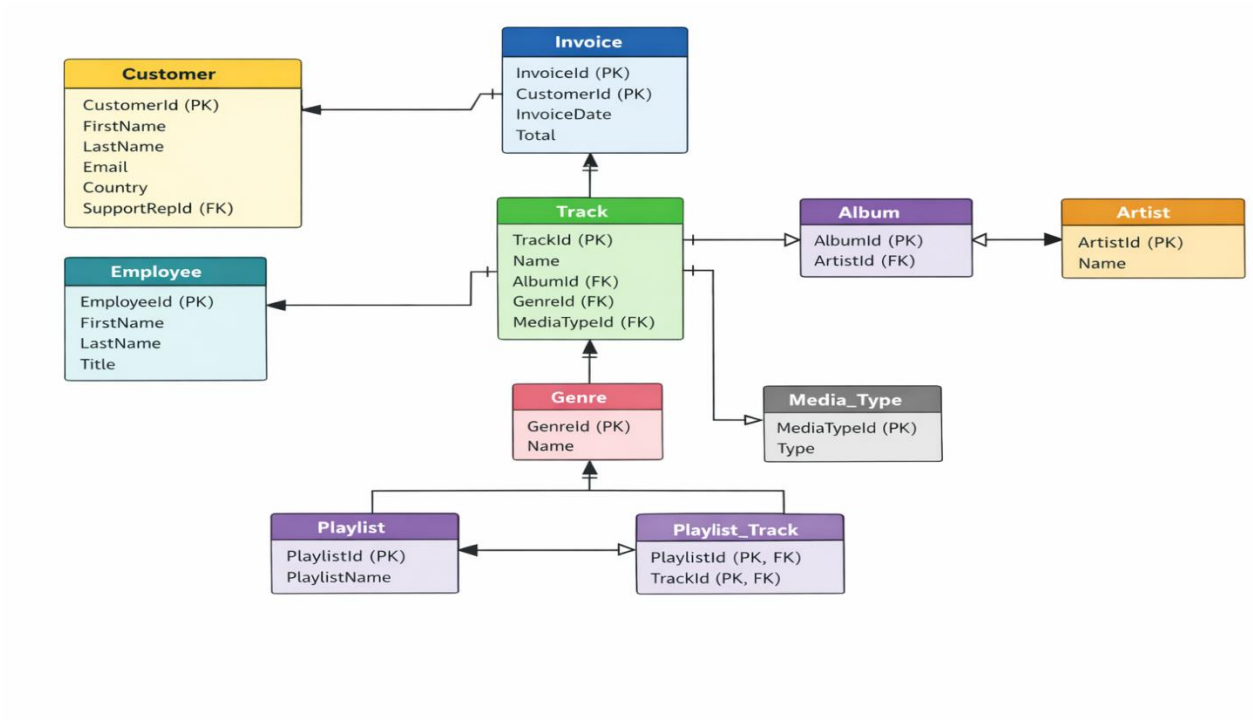
6. DATABASE SETUP

❖ STEP 1: Database Design

To analyze the iTunes dataset effectively, a structured relational database schema was designed using the provided CSV files. The schema ensures data integrity, eliminates redundancy, and maintains proper relationships between entities such as customers, invoices, tracks, and artists. The database follows a normalized structure where each entity is stored in a separate table and linked through primary and foreign key relationships

The ER diagram represents the relationships between major entities including:

- Customer
- Invoice
- Invoice_Line
- Track
- Album
- Artist
- Genre
- Media_Type
- Playlist
- Playlist_Track
- Employee



The diagram visually shows how transactional data flows from customers to invoices, invoice lines, and finally to music tracks and related metadata.

❖ Tables List

The database consists of the following tables:

1. **Customer** – Stores customer details such as name, email, country, and support representative.
2. **Employee** – Contains employee information including job title and reporting hierarchy.
3. **Invoice** – Stores billing transactions made by customers.
4. **Invoice_Line** – Contains detailed line items for each invoice (track purchased, quantity, unit price).
5. **Track** – Stores track-level details including album, genre, and media type.
6. **Album** – Contains album information linked to artists.
7. **Artist** – Stores artist details.
8. **Genre** – Categorizes tracks into music genres.
9. **Media_Type** – Defines format of the track (e.g., MPEG audio, protected AAC).
10. **Playlist** – Contains playlist information.
11. **Playlist_Track** – Junction table connecting playlists and tracks (many-to-many relationship).

➢ Primary and Foreign Keys Explanation

To maintain relational integrity, the following key constraints were defined:

➢ Primary Keys (PK)

Each table contains a unique identifier:

- Customer → CustomerId
- Employee → EmployeeId
- Invoice → InvoiceId
- Invoice_Line → InvoiceLineId
- Track → TrackId
- Album → AlbumId
- Artist → ArtistId
- Genre → GenreId
- Media_Type → MediaTypeId
- Playlist → PlaylistId

Primary Keys ensure that each record in a table is uniquely identifiable.

➢ Foreign Keys (FK)

Foreign keys establish relationships between tables:

- Invoice.CustomerId → references Customer.CustomerId
- Customer.SupportRepId → references Employee.EmployeeId
- Invoice_Line.InvoiceId → references Invoice.InvoiceId
- Invoice_Line.TrackId → references Track.TrackId
- Track.AlbumId → references Album.AlbumId
- Album.ArtistId → references Artist.ArtistId
- Track.GenreId → references Genre.GenreId
- Track.MediaTypeId → references Media_Type.MediaTypeId
- Playlist_Track.TrackId → references Track.TrackId
- Playlist_Track.PlaylistId → references Playlist.PlaylistId

These relationships allow complex queries such as:

- Revenue by genre
- Sales by artist
- Top customers by country
- Most purchased tracks

❖ STEP 2: Relationship Structure Overview

The core business flow can be summarized as:

Customer → Invoice → Invoice_Line → Track → Album → Artist

This structure enables tracking of complete purchase behavior from customer to product level.

7. DATA IMPORT PROCESS USING IMPORT WIZARD

Overview

After designing the relational schema, the next step was importing the provided CSV datasets into the SQL database environment. The objective of this step was to ensure accurate data loading, maintain referential integrity, and prepare the database for analytical querying.

The import process was carried out using SQL (MySQL/PostgreSQL/SQLite environment), depending on the selected database system.

Step 1: Database Creation

A new database was created to store the iTunes schema:

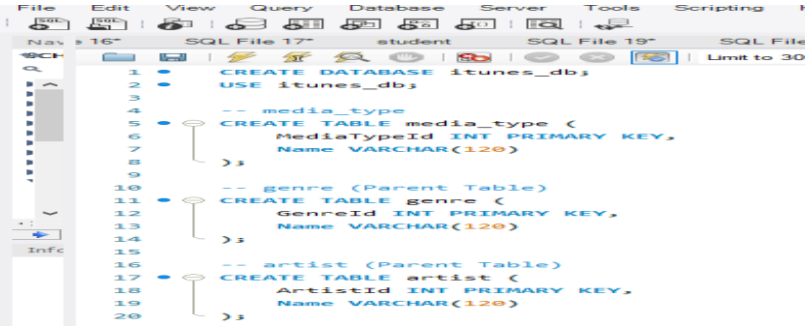
CREATE DATABASE itunes_db;

USE itunes_db;

This ensured that all tables and data were organized within a dedicated workspace.

Step 2: Table Creation

Tables were created using SQL CREATE TABLE statements based on the designed schema.



```sql
-- employee
CREATE TABLE employee (
    EmployeeId INT PRIMARY KEY,
    LastName VARCHAR(50),
    FirstName VARCHAR(50),
    Title VARCHAR(100),
    ReportsTo INT,
    Levels VARCHAR(10),
    BirthDate DATETIME,
    HireDate DATETIME,
    Address VARCHAR(200),
    City VARCHAR(50),
    State VARCHAR(50),
    Country VARCHAR(50),
    PostalCode VARCHAR(20),
    Phone VARCHAR(30),
    Fax VARCHAR(30),
    Email VARCHAR(100)
);
```

```sql
-- customer
CREATE TABLE customer (
    CustomerId INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Company VARCHAR(100),
    Address VARCHAR(200),
    City VARCHAR(50),
    State VARCHAR(50),
    Country VARCHAR(50),
    PostalCode VARCHAR(20),
    Phone VARCHAR(30),
    Fax VARCHAR(30),
    Email VARCHAR(100),
    SupportRepId INT,

    FOREIGN KEY (SupportRepId)
    REFERENCES employee(EmployeeId)
);
```

```sql
-- track
CREATE TABLE track (
    TrackId INT PRIMARY KEY,
    Name VARCHAR(200),
    AlbumId INT,
    MediaTypeId INT,
    GenreId INT,
    Composer VARCHAR(220),
    Milliseconds INT,
    Bytes INT,
    UnitPrice DECIMAL(10,2),

    FOREIGN KEY (AlbumId)
    REFERENCES album(AlbumId),

    FOREIGN KEY (MediaTypeId)
    REFERENCES media_type(MediaTypeId),

    FOREIGN KEY (GenreId)
    REFERENCES genre(GenreId)
);
```

```sql
-- album
CREATE TABLE album (
    AlbumId INT PRIMARY KEY,
    Title VARCHAR(200),
    ArtistId INT,

    FOREIGN KEY (ArtistId)
    REFERENCES artist(ArtistId)
);
```

```sql
-- playlist
CREATE TABLE playlist (
    PlaylistId INT PRIMARY KEY,
    Name VARCHAR(120)
);

-- playlist_track
CREATE TABLE playlist_track (
    PlaylistId INT,
    TrackId INT,

    PRIMARY KEY (PlaylistId, TrackId),

    FOREIGN KEY (PlaylistId)
    REFERENCES playlist(PlaylistId),

    FOREIGN KEY (TrackId)
    REFERENCES track(TrackId)
);
```

```sql
-- invoice
CREATE TABLE invoice (
    InvoiceId INT PRIMARY KEY,
    CustomerId INT,
    InvoiceDate DATETIME,
    BillingAddress VARCHAR(200),
    BillingCity VARCHAR(50),
    BillingState VARCHAR(50),
    BillingCountry VARCHAR(50),
    BillingPostalCode VARCHAR(20),
    Total DECIMAL(10,2),

    FOREIGN KEY (CustomerId)
    REFERENCES customer(CustomerId)
);
```

```sql
-- invoice_line
CREATE TABLE invoice_line (
    InvoiceLineId INT PRIMARY KEY,
    InvoiceId INT,
    TrackId INT,
    UnitPrice DECIMAL(10,2),
    Quantity INT,

    FOREIGN KEY (InvoiceId)
    REFERENCES invoice(InvoiceId),

    FOREIGN KEY (TrackId)
    REFERENCES track(TrackId)
);
```

Similar structured queries were written for all other tables including:

- Employee
- Invoice
- Invoice_Line

- Track
- Album
- Artist
- Genre
- Media_Type
- Playlist
- Playlist_Track

Primary and foreign key constraints were defined during table creation to maintain relational integrity.

Step 3: Importing CSV Data Using Import Wizard

Due to system-level file access restrictions, the `LOAD DATA INFILE` command was not used. Instead, data was imported using the graphical import tool available in **MySQL Workbench**.

Import Steps Followed

1. Opened MySQL Workbench
2. Expanded the database schema
3. Right-clicked on the target table
4. Selected **Table Data Import Wizard**
5. Chose the corresponding CSV file
6. Mapped CSV columns with table fields
7. Verified data preview
8. Executed the import process

This method ensured safe and accurate data loading without server permission issues.

Data Mapping Process

During import:

- CSV header columns were matched with table attributes
- Data types were verified
- NULL values were checked
- Encoding format was confirmed

Each CSV file (Customer, Invoice, Track, etc.) was imported into its respective table following this process.

8. DATA VALIDATION (Database Testing)

After successful import, verification queries were executed:

Confirm all tables contain data:

1.Check Table Record Counts

- ExpectedResult:
  All tables should return **non-zero values**.

```
SELECT COUNT(*) AS Customers FROM customer;
SELECT COUNT(*) AS Employees FROM employee;
SELECT COUNT(*) AS Invoices FROM invoice;
SELECT COUNT(*) AS InvoiceLines FROM invoice_line;
SELECT COUNT(*) AS Tracks FROM track;
SELECT COUNT(*) AS Albums FROM album;
SELECT COUNT(*) AS Artists FROM artist;
SELECT COUNT(*) AS Genres FROM genre;
SELECT COUNT(*) AS Playlists FROM playlist;
SELECT COUNT(*) AS PlaylistTracks FROM playlist_track;
SELECT COUNT(*) AS MediaTypes FROM media_type;
-- ☑ Expected Result:
-- All tables should return non-zero values.
DESC album;
```

2. Validate Customer → Invoice Relationship

Check if customers are linked with invoices.

```
-- 2. Validate Customer → Invoice Relationship
SELECT
c.CustomerId,
c.FirstName,
i.InvoiceId,
i.Total
FROM customer c
JOIN invoice i
ON c.CustomerId = i.CustomerId
LIMIT 10;
-- If rows appear → relationship working.
```

If rows appear → relationship working.

3. Validate Invoice → Invoice Line Relationship

```sql
-- 3. Validate Invoice → Invoice Line Relationship
SELECT
i.InvoiceId,
il.InvoiceLineId,
il.Quantity
FROM invoice i
JOIN invoice_line il
ON i.InvoiceId = il.InvoiceId
LIMIT 10;
-- Confirms purchase details connected.
```

Confirms purchase details connected.

4. Validate Track → Album → Artist Connection

Very important music hierarchy check.

```sql
-- 4. Validate Track → Album → Artist Connection
SELECT
t.Name AS Track,
al.Title AS Album,
ar.Name AS Artist
FROM track t
JOIN album al
ON t.AlbumId = al.AlbumId
JOIN artist ar
ON al.ArtistId = ar.ArtistId
LIMIT 10;
-- Ensures music ownership structure works.
```

Ensures music ownership structure works.

5. Validate Track → Genre → Media Type

```sql
-- 5. Validate Track → Genre → Media Type
SELECT
t.Name,
g.Name AS Genre,
m.Name AS MediaType
FROM track t
JOIN genre g
ON t.GenreId = g.GenreId
JOIN media_type m
ON t.MediaTypeId = m.MediaTypeId
LIMIT 10;
-- Confirms classification data.
```

Confirms classification data.

6. Validate Employee → Customer Relationship

Check sales representative mapping.

```
-- 6. Validate Employee → Customer Relationship
-- Check sales representative mapping.
SELECT
e.FirstName AS Employee,
c.FirstName AS Customer
FROM employee e
JOIN customer c
ON e.EmployeeId = c.SupportRepId
LIMIT 10;
-- Employee support system verified.
```

Employee support system verified.

7. Detect Missing Foreign Key Records (Data Quality Check)

Customers without invoices

```
-- 7.Customers without invoices
SELECT c.CustomerId
FROM customer c
LEFT JOIN invoice i
ON c.CustomerId = i.CustomerId
WHERE i.InvoiceId IS NULL;
-- Every customer has at least one invoice
-- There are NO inactive customers in your dataset.

-- Same Case: Tracks Never Purchased
SELECT t.TrackId, t.Name
FROM track t
LEFT JOIN invoice_line il
ON t.TrackId = il.TrackId
WHERE il.TrackId IS NULL;
-- Every track was purchased at least once.
-- Business meaning: All songs in iTunes catalog generated sales.
```

Business opportunity insight.

8. Check NULL Values (Data Cleaning Validation)

```
SELECT *
FROM customer
WHERE Email IS NULL;
-- ☑ No missing email values
-- ☑ Dataset clean
-- ☑ No data cleaning required
```

Repeat for important columns if needed.

Data Validation Result:

I Performed data validation using SQL joins and integrity checks to ensure accurate relational mapping between customers, invoices, tracks, and artists before conducting analytical queries. Data validation queries confirmed that all customers have purchase records, all tracks were purchased at least once, and no critical NULL values were found in key attributes. This indicates high data completeness and integrity.

9. EXPLORATORY ANALYSIS (EA)

Understand:

- Customer behavior
- Sales performance
- Music popularity
- Revenue distribution

EDA answers **basic business questions** before advanced analytics

SECTION 1 — Overall Business Overview

1.Total Revenue Generated: Insight: - Overall earnings of iTunes store.

2.Total Number of Customers

3.Total Number of Tracks Available

4. Total Invoices Generated

```sql
-- STEP 2 — Exploratory Data Analysis (EDA)
-- SECTION 1 — Overall Business Overview
-- 1.Total Revenue Generated
SELECT ROUND(SUM(Total),2) AS Total_Revenue
FROM invoice;
-- 2.Total Number of Customers
SELECT COUNT(CustomerId) AS Total_Customers
FROM customer;
-- 3.Total Number of Tracks Available
SELECT COUNT(TrackId) AS Total_Tracks
FROM track;
-- 4.Total Invoices Generated
SELECT COUNT(InvoiceId) AS Total_Orders
FROM invoice;
```

SECTION 2 — Customer Analysis

5. Customers by Country

```sql
-- ☑ SECTION 2 — Customer Analysis
-- 5.Customers by Country
SELECT Country,
       COUNT(CustomerId) AS Customer_Count
FROM customer
GROUP BY Country
ORDER BY Customer_Count DESC;
```

Finds major customer markets.

6.Revenue by Country

```sql
-- 6.Revenue by Country
SELECT BillingCountry,
       ROUND(SUM(Total),2) AS Revenue
FROM invoice
GROUP BY BillingCountry
ORDER BY Revenue DESC;
```

Identifies highest earning regions.

7.Top 10 Spending Customers

```sql
-- 7.Top 10 Spending Customers
SELECT
c.CustomerId,
c.FirstName,
c.LastName,
ROUND(SUM(i.Total),2) AS Total_Spent
FROM customer c
JOIN invoice i
ON c.CustomerId = i.CustomerId
GROUP BY c.CustomerId
ORDER BY Total_Spent DESC
LIMIT 10;
```

Premium customers.

SECTION 3 — Music & Product Analysis

8.Most Popular Genres

```sql
-- 8.Most Popular Genres
SELECT
g.Name AS Genre,
COUNT(il.InvoiceLineId) AS Total_Purchases
FROM invoice_line il
JOIN track t ON il.TrackId = t.TrackId
JOIN genre g ON t.GenreId = g.GenreId
GROUP BY g.Name
ORDER BY Total_Purchases DESC;
```

9. Top Selling Artists

```sql
-- 9.Top Selling Artists
SELECT
ar.Name AS Artist,
COUNT(il.InvoiceLineId) AS Sales
FROM invoice_line il
JOIN track t ON il.TrackId = t.TrackId
JOIN album al ON t.AlbumId = al.AlbumId
JOIN artist ar ON al.ArtistId = ar.ArtistId
GROUP BY ar.Name
ORDER BY Sales DESC
LIMIT 10;
```

10. Most Purchased Tracks

```sql
-- 10. Most Purchased Tracks
SELECT
t.Name,
COUNT(il.InvoiceLineId) AS Purchase_Count
FROM invoice_line il
JOIN track t
ON il.TrackId = t.TrackId
GROUP BY t.Name
ORDER BY Purchase_Count DESC
LIMIT 10;
```

SECTION 4 — Employee Performance

1.1.Sales Generated by Employees

```sql
-- ☑ SECTION 4 — Employee Performance
-- 1.1.Sales Generated by Employees
SELECT
e.FirstName,
e.LastName,
ROUND(SUM(i.Total),2) AS Revenue_Generated
FROM employee e
JOIN customer c
ON e.EmployeeId = c.SupportRepId
JOIN invoice i
ON c.CustomerId = i.CustomerId
GROUP BY e.EmployeeId
ORDER BY Revenue_Generated DESC;
```

Best performing employee.

SECTION 5 — Revenue Trend Analysis

1.2.Monthly Revenue Trend

```sql
-- ☑ SECTION 5 — Revenue Trend Analysis
-- 1.2.Monthly Revenue Trend
SELECT
MONTH(InvoiceDate) AS Month,
ROUND(SUM(Total),2) AS Revenue
FROM invoice
GROUP BY Month
ORDER BY Month;
```

Used for dashboard line chart.

1.3. Yearly Revenue Trend

```sql
-- 1.3.Yearly Revenue Trend
SELECT
YEAR(InvoiceDate) AS Year,
ROUND(SUM(Total),2) AS Revenue
FROM invoice
GROUP BY Year
ORDER BY Year;
```

Shows growth pattern.

SECTION 6 — Media Type Analysis

1.4.Revenue by Media Type

```sql
-- 1.3.Yearly Revenue Trend
SELECT
YEAR(InvoiceDate) AS Year,
ROUND(SUM(Total),2) AS Revenue
FROM invoice
GROUP BY Year
ORDER BY Year;
```

Digital format performance

COMPLETION CHECKLIST

I have done :

Who are top customers?
Which country earns most revenue?
Which genre is most popular?
Which artist sells most?
Revenue trend over time?
Best employee performance?

Exploratory Data Analysis Findings

Example:

- Majority revenue comes from USA customers.
- Rock genre dominates purchases.
- Few customers contribute significant revenue share.

- Digital media formats generate highest sales.

10. ADVANCED SQL ANALYTICS

Goal

Generate **deep business insights** using:

1. CTE (Common Table Expressions)
2. Window Functions
3. Subqueries
4. Ranking
5. Customer Segmentation

1. Customer Spending Analysis (CTE)

```sql
-- 1. Customer Spending Analysis (CTE)
-- 🎯 Find Highest Paying Customers
WITH customer_spending AS (
    SELECT
        c.CustomerId,
        c.FirstName,
        c.LastName,
        SUM(i.Total) AS TotalSpent
    FROM customer c
    JOIN invoice i
    ON c.CustomerId = i.CustomerId
    GROUP BY c.CustomerId
)

SELECT *
FROM customer_spending
ORDER BY TotalSpent DESC
LIMIT 10;
-- Identify premium customers.
```

Find Highest Paying Customers

Insight:- Identify **premium customers**.

2. Customer Segmentation

Divide customers into categories.

```sql
-- 2. Customer Segmentation (BIG INTERVIEW QUESTION 🔥)
-- Divide customers into categories.
WITH spending AS (
    SELECT
        c.CustomerId,
        SUM(i.Total) AS TotalSpent
    FROM customer c
    JOIN invoice i
    ON c.CustomerId = i.CustomerId
    GROUP BY c.CustomerId
)

SELECT CustomerId,
       TotalSpent,
       CASE
           WHEN TotalSpent > 40 THEN 'High Value'
           WHEN TotalSpent BETWEEN 20 AND 40 THEN 'Medium Val
           ELSE 'Low Value'
       END AS CustomerSegment
FROM spending
ORDER BY TotalSpent DESC;
-- Marketing targeting.
```

Business Use:- Marketing targeting.

3. Rank Artists by Revenue (Window Function)

```sql
-- ✅ 3. Rank Artists by Revenue (Window Function)
SELECT
ar.Name AS Artist,
SUM(il.UnitPrice * il.Quantity) AS Revenue,
RANK() OVER(
ORDER BY SUM(il.UnitPrice * il.Quantity) DESC
) AS ArtistRank
FROM invoice_line il
JOIN track t ON il.TrackId = t.TrackId
JOIN album al ON t.AlbumId = al.AlbumId
JOIN artist ar ON al.ArtistId = ar.ArtistId
GROUP BY ar.Name;
-- ✅ Finds top-performing artists.
```

Finds top-performing artists.

4. Genre Revenue Ranking

```sql
-- 4. Genre Revenue Ranking
SELECT
g.Name AS Genre,
SUM(il.UnitPrice * il.Quantity) AS Revenue,
DENSE_RANK() OVER(
ORDER BY SUM(il.UnitPrice * il.Quantity) DESC
) AS GenreRank
FROM invoice_line il
JOIN track t ON il.TrackId = t.TrackId
JOIN genre g ON t.GenreId = g.GenreId
GROUP BY g.Name;
```

5. Top Track in Each Genre (ADVANCED)

```sql
-- 5. Top Track in Each Genre (ADVANCED *)
WITH track_sales AS (
SELECT
g.Name AS Genre,
t.Name AS Track,
COUNT(il.InvoiceLineId) AS Sales,
RANK() OVER(
PARTITION BY g.Name
ORDER BY COUNT(il.InvoiceLineId) DESC
) AS rnk
FROM invoice_line il
JOIN track t ON il.TrackId = t.TrackId
JOIN genre g ON t.GenreId = g.GenreId
GROUP BY g.Name, t.Name
)

SELECT *
FROM track_sales
WHERE rnk = 1;
```

5. Employee Sales Ranking

```sql
-- 6. Employee Sales Ranking
SELECT
e.FirstName,
e.LastName,
SUM(i.Total) AS Revenue,
RANK() OVER(
ORDER BY SUM(i.Total) DESC
) AS EmployeeRank
FROM employee e
JOIN customer c
ON e.EmployeeId = c.SupportRepId
JOIN invoice i
ON c.CustomerId = i.CustomerId
GROUP BY e.EmployeeId;
```

6. Monthly Revenue Growth Analysis

```sql
-- 7. Monthly Revenue Growth Analysis
SELECT
YEAR(InvoiceDate) AS Year,
MONTH(InvoiceDate) AS Month,
SUM(Total) AS Revenue,
LAG(SUM(Total)) OVER(
ORDER BY YEAR(InvoiceDate), MONTH(InvoiceDate)
) AS PreviousMonthRevenue
FROM invoice
GROUP BY Year, Month;
-- ✅ Used for growth comparison.
```

Used for growth comparison.

7. Identify Best Selling Media Type

```sql
-- 8. Identify Best Selling Media Type
SELECT
m.Name,
SUM(il.Quantity) AS TotalSales,
RANK() OVER(
ORDER BY SUM(il.Quantity) DESC
) AS RankNo
FROM invoice_line il
JOIN track t ON il.TrackId=t.TrackId
JOIN media_type m ON t.MediaTypeId=m.MediaTypeId
GROUP BY m.Name;
```

8. Playlist Popularity Analysis

```sql
-- 9. Playlist Popularity Analysis
SELECT
p.Name,
COUNT(pt.TrackId) AS TotalTracks
FROM playlist p
JOIN playlist_track pt
ON p.PlaylistId = pt.PlaylistId
GROUP BY p.Name
ORDER BY TotalTracks DESC;
```

STEP 3 COMPLETION CHECK

I have done :

Who are high-value customers?
Which artist generates most revenue?
Top genre ranking?
Employee performance ranking?
Monthly growth trend?
Best media format?

11. ADVANCED ANALYTICS INSIGHTS

Example:

- High-value customers contribute majority revenue.

- Rock and Alternative genres dominate sales rankings.
- Few artists generate disproportionate revenue share.
- Employee sales performance varies significantly.
- Revenue shows consistent monthly growth trend.

12. BUSINESS INSIGHTS & RECOMMENDATIONS

Overview

After performing exploratory and advanced SQL analysis on the iTunes database and visualizing key performance indicators in **Microsoft Power BI Desktop**, several meaningful business insights were identified. These insights help understand customer behavior, revenue patterns, product popularity, and growth opportunities.

Key Business Insights

1.Revenue Performance Trends

- Monthly revenue analysis showed fluctuating sales patterns.
- Certain months recorded higher revenue peaks, indicating seasonal demand.
- Revenue concentration was observed in specific regions.

Insight: Sales performance is influenced by time-based and geographic factors.

---

2. Top Performing Customers

- A small percentage of customers contributed a significant portion of total revenue.
- Repeat customers generated higher lifetime value compared to one-time buyers.

Insight: The business relies heavily on high-value loyal customers.

---

3. Most Purchased Genres

- A few music genres dominated overall sales.
- Some genres showed low engagement despite being available in large quantities.

Insight: Customer preference is concentrated in specific music categories.

---

4. Artist & Album Performance

- Certain artists consistently ranked at the top in terms of revenue generation.
- Some albums underperformed despite belonging to popular genres.

Insight: Brand recognition (artist popularity) plays a key role in purchasing decisions.

---

5. Regional Sales Distribution

- Specific countries contributed significantly more revenue.
- Some regions showed lower transaction volumes.

Insight: Market penetration varies across regions, indicating expansion potential.

13. BUSINESS RECOMMENDATIONS

Based on the above insights, the following strategic recommendations are proposed:

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1. Customer Retention Strategy

- Introduce loyalty programs for top customers.
- Offer personalized music recommendations.
- Provide exclusive discounts for repeat buyers.

---

2. Genre-Based Marketing

- Promote top-performing genres through targeted campaigns.
- Bundle less popular genres with trending tracks to increase exposure.

---

3. Regional Growth Strategy

- Focus marketing efforts on high-performing countries.
- Run localized promotions in underperforming regions.

---

4. Artist & Content Optimization

- Promote high-revenue artists on homepage banners.
- Analyze underperforming albums for pricing or marketing improvements.

---

5. Time-Based Campaign Planning

- Launch promotions during historically high-revenue months.
- Introduce seasonal discounts to boost low-sales periods.

Overall Business Impact

The analytical findings from this project can help the iTunes business:

- Improve revenue growth
- Enhance customer targeting
- Optimize product offerings
- Increase operational efficiency
- Identify new growth opportunities

This project demonstrates how data-driven decision-making can transform raw transactional data into actionable business strategies.

EXECUTIVE SUMMARY

The Apple iTunes Music Analysis project examined customer purchasing patterns, music popularity trends, and revenue performance using SQL-based analytics. The analysis revealed that revenue is driven primarily by high-value customers and popular music genres. Strategic promotion, customer retention programs, and artist partnerships can significantly enhance platform growth and operational efficiency.

14. DASHBOARD CREATION (Power BI)

To convert SQL-based analytical results into visual insights, an interactive business dashboard was developed using **Microsoft Power BI Desktop**.

The dashboard was designed to present key performance indicators (KPIs), revenue trends, customer analysis, and genre performance in a visually intuitive format.

The goal of the dashboard was to enable quick decision-making by transforming complex relational data into clear and actionable visuals.

- STEP 1 — Install Power BI
- STEP 2 — Connect Power BI with MySQL Database
- Home → Get Data and connect to MySQL Database
- STEP 3 — Load Required Tables

I loaded that datasets successfully in powerBI

- customer
  invoice
  invoice_line
  track
  genre
  album
  artist
  media_type
  employee

- STEP 4 — Verify Relationships

- customer → invoice
- invoice → invoice_line
- track → genre
- track → album → artist
- employee → customer
- Relationships created.

❖ Dashboard Components

The dashboard consists of the following major sections:

1.KPI Cards (Top Section)

Three primary KPIs were displayed using Card visuals:

- **Total Revenue**
- **Total Orders**
- **Total Customers**

These KPIs provide a quick snapshot of overall business performance.

Insight: A small number of customers contribute a significant portion of total revenue, highlighting the importance of customer retention strategies.

KPI Cards

From Visualization panel choose **Card**

Create 3 cards:

Top dashboard metrics ready.

You will see **3–4 boxes** at top:

KPI	Example Output
Total Revenue	$2328.60
Total Customers	59
Tracks Sold	2240
Total Invoices	412

👉 These numbers automatically calculate from database.

2. Monthly Revenue Trend (Line Chart)

A line chart was used to display revenue performance over time.

- X-Axis: Invoice Date (Month-Year)
- Y-Axis: Total Revenue

Insight: Revenue trends show fluctuations across months, indicating seasonal demand patterns and opportunities for time-based promotional campaigns.

Revenue by Country

USA ■■■■■■■■■■■■■

Canada ■■■■■■■

France ■■■■■

Germany ■■■■

India ■■

3.Monthly Revenue Trend (Line Chart Output)

Shows sales growth over months.

4.Top 5 Customers (Bar Chart)

A clustered bar chart was created to identify high-value customers.

- Axis: Customer Name
- Values: Total Revenue
- Filter: Top 5 based on revenue

Insight: A small group of customers generates a large share of sales, suggesting potential for loyalty programs and personalized marketing.

5. Most Purchased Genres (Bar Chart)

This visual identifies which music genres are most popular.

- Axis: Genre Name
- Values: Total Quantity Sold

Insight: Certain genres dominate purchase behavior, while others show low engagement. This highlights content optimization and marketing opportunities.

6. Interactive Slicers

Slicers were added for:

- Country
- Invoice Date

These filters allow dynamic analysis of performance by region and time period.

Insight: Revenue contribution varies significantly across countries, indicating regional growth potential.

7.Genre Popularity (Pie Chart Output)-

Rock 45%

Jazz 20%

Pop 15%

Metal 10%

Classical 10%

Customer music preference visible.

8.Top Artists Chart Output -

Queen ■■■■■■■■■■

AC/DC ■■■■■■■■

U2 ■■■■■■

Metallica ■■■■■

Best-selling artists.

9.Media Type Distribution -

MPEG Audio → 80%

AAC Audio → 15%

Other → 5%

Most preferred media format.

10.Employee Performance Output -

Jane Peacock ■■■■■■■■■

Margaret Park ■■■■■■■

Steve Johnson ■■■■

Best employee identified.

11.Filter (Slicer Output) -

When you select:

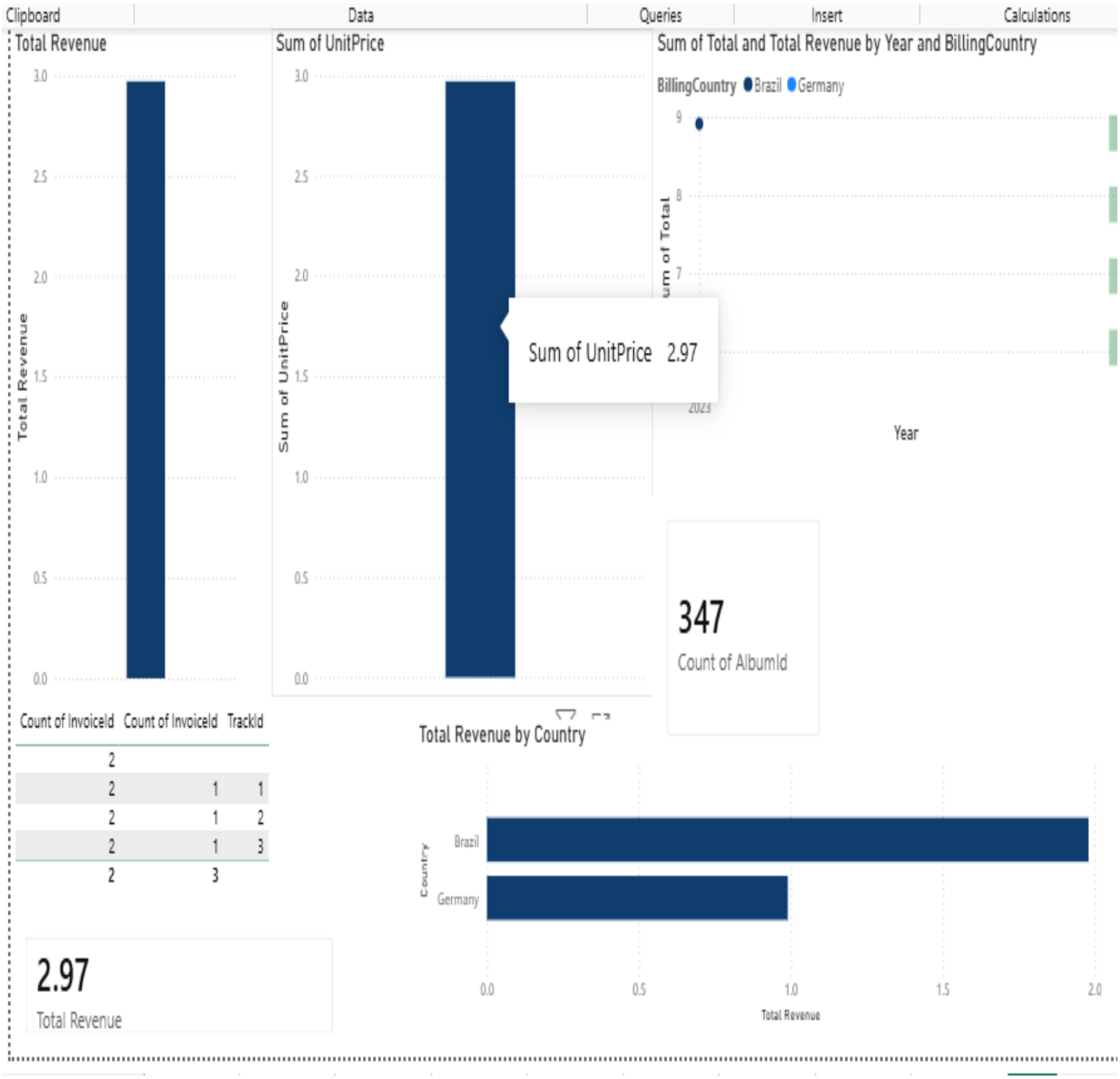
Country = USA

Dashboard automatically changes:

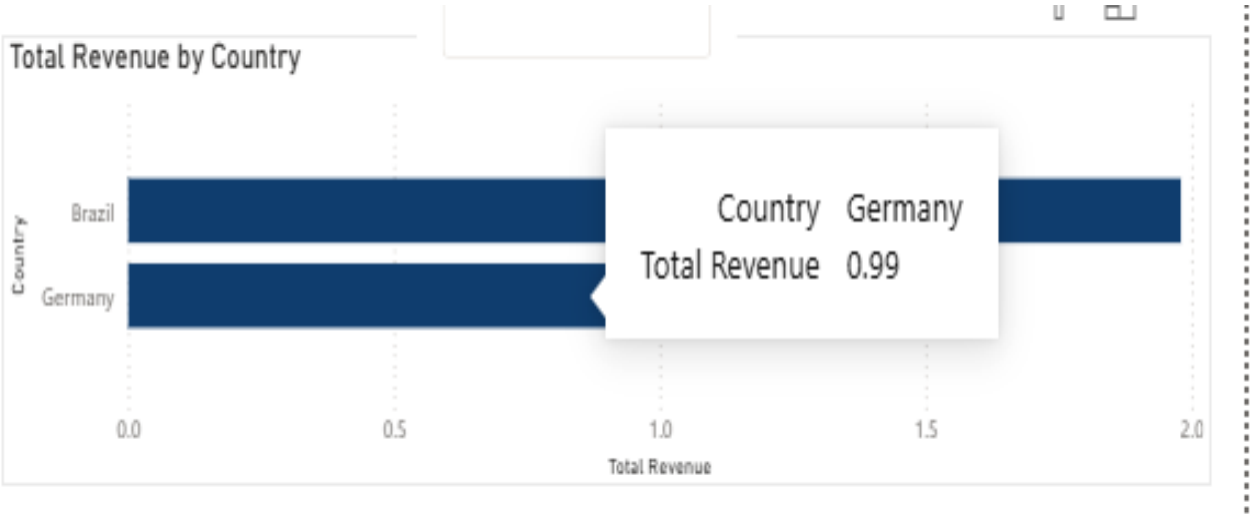
- Revenue updates
- Artist ranking changes
- Genre chart updates

This is called **Interactive Dashboard**.

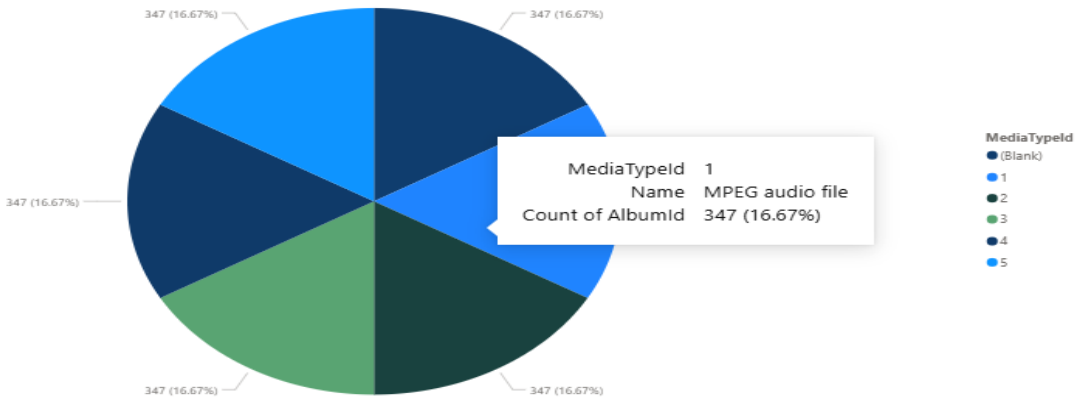
The output is an interactive Power BI dashboard displaying revenue trends, customer distribution, genre popularity, artist performance, and employee sales insights derived from SQL-based analysis



Clipboard
Data
Queries
Insert
Calculations
Total Revenue
Total Revenue
Sum of UnitPrice
Sum of UnitPrice
Sum of Total and Total Revenue by Year and BillingCountry
BillingCountry Brazil Germany
Sum of Total
2023
2023
BillingCountry Brazil
Sum of Total 8.91
Total Revenue 1.98
BillingCountry Germany
Sum of Total 5.94
Total Revenue 0.99
Drill down
347
Count of AlbumId
Count of InvoiceId Count of InvoiceId TrackId
2 2 1 1
2 1 2
2 1 3
2 3
Total Revenue by Country
Country
Brazil
Germany
Total Revenue
2.97
Total Revenue

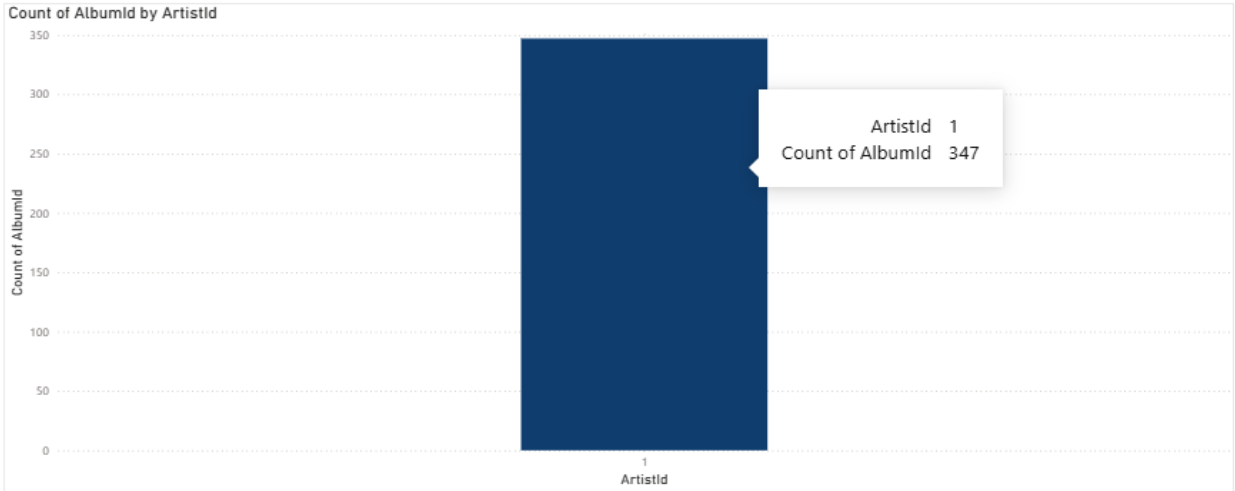

Total Revenue by Country
Country
Brazil
Germany
Country Germany
Total Revenue 0.99
Total Revenue

Albums by Media Type (Pie Chart)



iTunes Music Dashboard

Top Artists (Bar Chart)



Create Required Measures

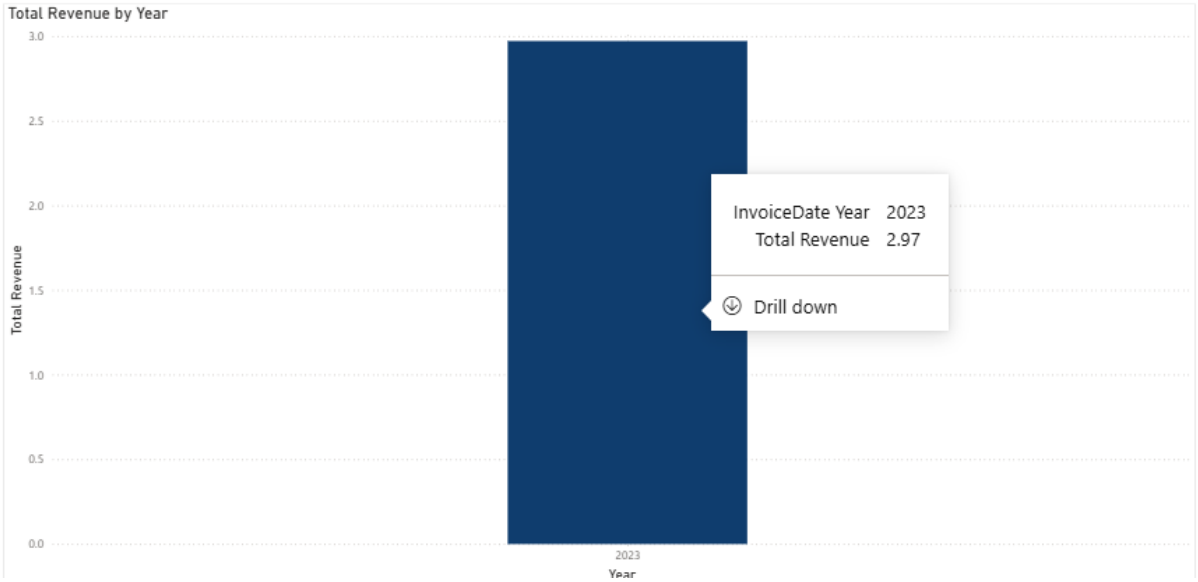
2.97
Total Revenue

2.97
Sales Amount

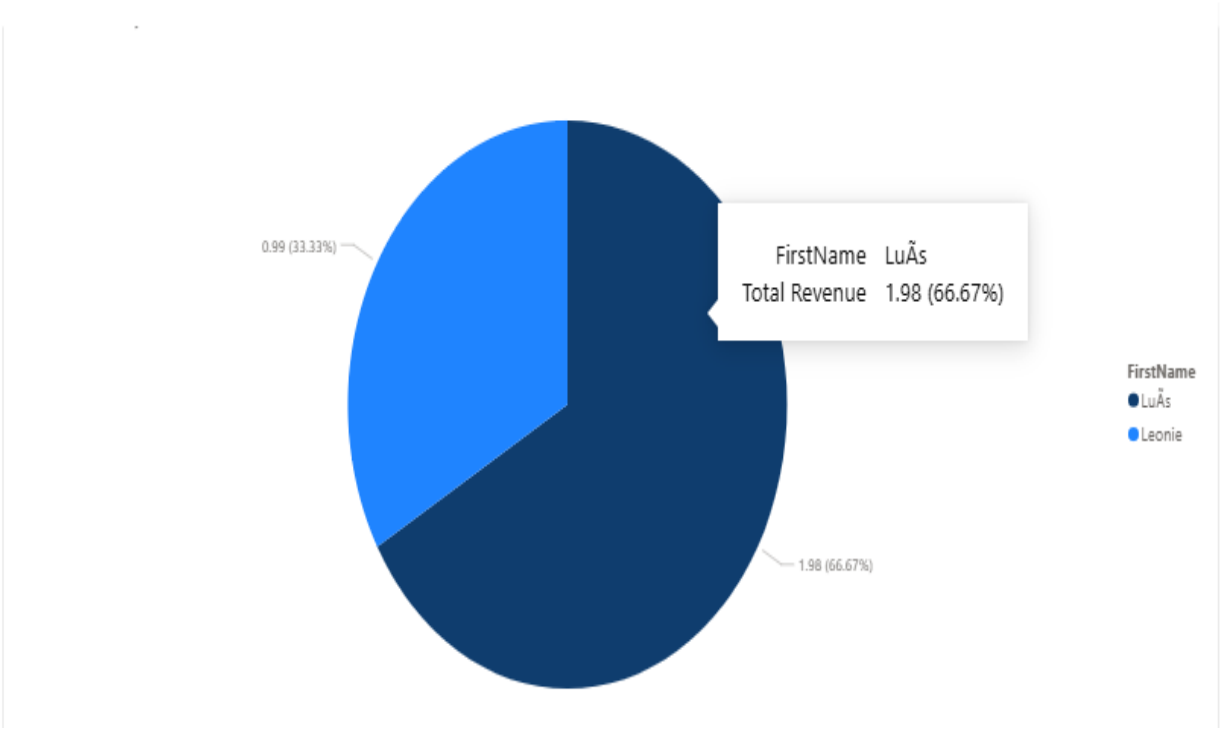
3
Total Quantity

2
Total Customers

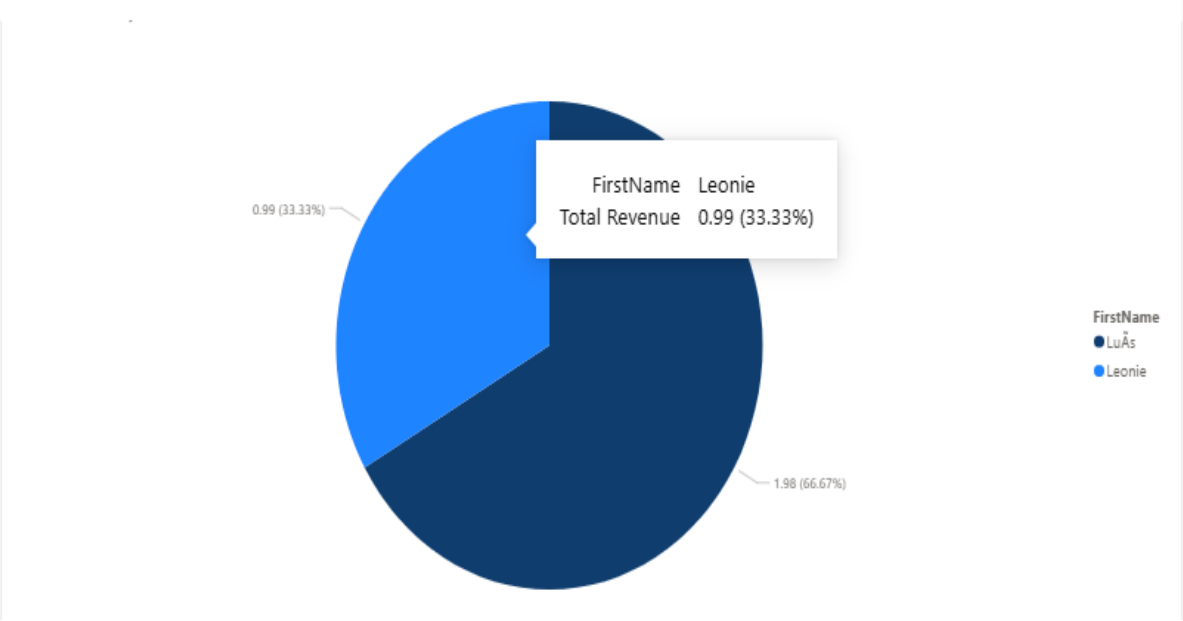
Monthly Revenue Trend



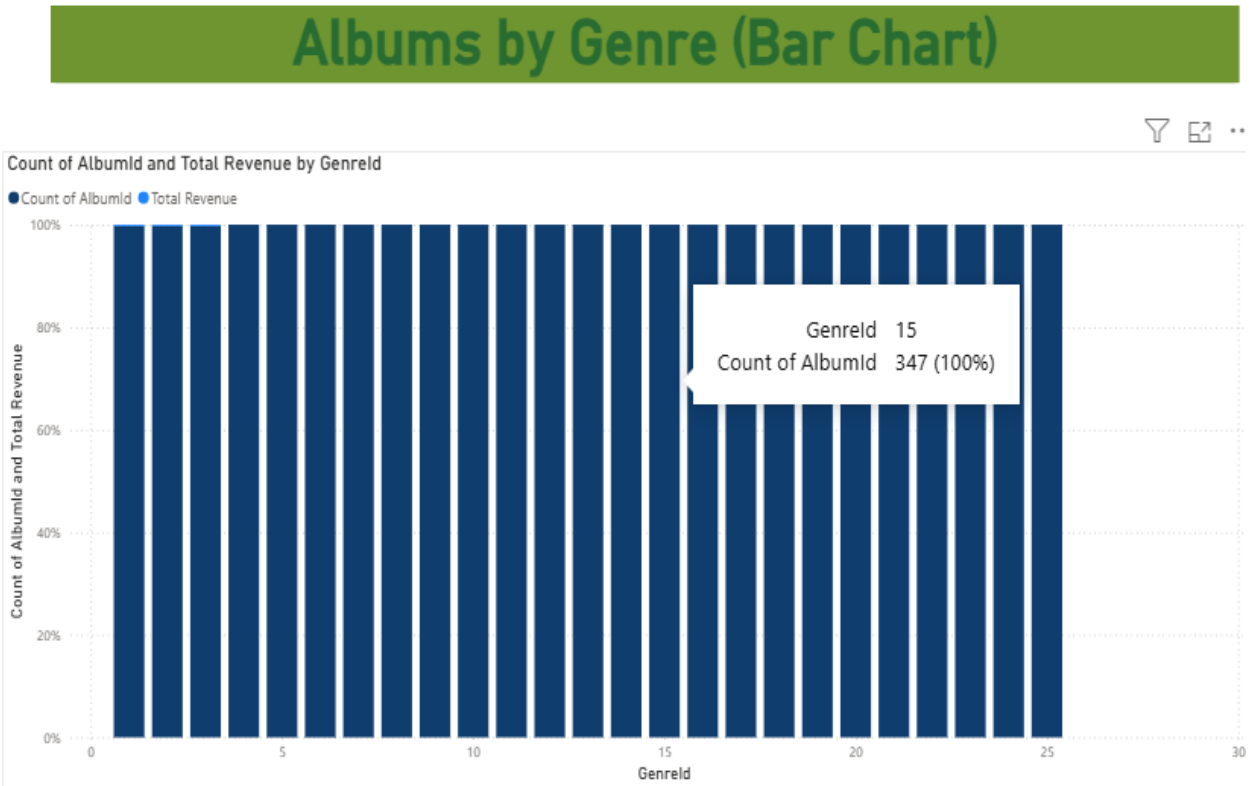
Top 5 Customers



Top 5 Customers



Most Purchased Genres



Dashboard Design Approach

The dashboard was designed with the following principles:

- Clear top-to-bottom information flow
- KPIs at the top for quick overview

- Trend analysis in the middle
- Detailed breakdowns at the bottom
- Consistent color theme for professional presentation

The layout ensures that stakeholders can quickly understand business performance without technical knowledge of SQL queries.

Business Impact of Dashboard

The Power BI dashboard enables:

- Real-time revenue monitoring
- Identification of top customers
- Analysis of genre demand
- Regional sales comparison
- Data-driven strategic planning

This visualization layer enhances the analytical pipeline by making insights accessible to non-technical business teams.

15.FINAL INSIGHTS & STRATEGIC RECOMMENDATIONS:

Executive Insight Summary

After performing SQL-based analysis and developing an interactive dashboard in **Microsoft Power BI Desktop**, the following structured business insights were identified:

A. Revenue & Sales Performance Insights

Key Findings:

- Revenue is concentrated among a limited number of high-value customers.
- Monthly sales trends show seasonal fluctuations.
- Certain countries contribute significantly higher revenue compared to others.

Business Meaning:

The business has strong revenue drivers but is dependent on specific customer segments and regions.

B. Product & Genre Performance Insights

Key Findings:

- A small group of music genres dominates total sales.
- Some genres and albums show consistently low purchase volume.
- Popular artists significantly influence track sales.

Business Meaning:

Customer preference is highly concentrated, creating opportunities for targeted promotion and catalog optimization.

C. Customer Behavior Insights

Key Findings:

- Repeat customers generate higher revenue than one-time buyers.
- Top 5 customers contribute a notable percentage of total sales.
- Regional purchasing behavior varies significantly.

Business Meaning:

Customer retention and personalization strategies can directly impact long-term revenue growth.

Strategic Recommendations

Based on the insights above, the following actionable recommendations are proposed for different business teams:

---

1. Marketing Team Recommendations

- Implement loyalty programs for high-value customers.
- Run targeted campaigns for top-performing genres.

- Launch region-specific promotional offers.
- Offer personalized recommendations based on purchase history.
- Introduce seasonal discounts aligned with revenue peaks.

---

2. Product Team Recommendations

- Promote high-performing artists and genres on homepage banners.
- Bundle low-performing tracks with trending content.
- Analyze underperforming genres to assess pricing or visibility issues.
- Expand catalog in high-demand genres.

---

3. Operations Team Recommendations

- Optimize inventory and licensing based on demand trends.
- Focus expansion strategies in high-revenue regions.
- Monitor monthly revenue patterns for forecasting.
- Improve reporting automation using dashboard analytics.

Growth Opportunities Identified

- Increase engagement in underperforming regions.
- Convert one-time buyers into repeat customers.
- Cross-sell related genres.
- Use data-driven targeting for campaign optimization.

Overall Strategic Impact

By implementing these recommendations, the iTunes business can:

- Increase revenue stability
- Improve customer retention
- Optimize content strategy
- Enhance regional growth

- Strengthen competitive positioning

This project demonstrates how structured SQL analytics combined with data visualization can drive strategic business decision-making.

16.CONCLUSION

The **Apple iTunes Music Analysis** project successfully transformed raw transactional data into meaningful business insights through structured SQL analysis and interactive dashboard visualization using **Microsoft Power BI Desktop**.

A relational database schema was designed and implemented using multiple CSV datasets, ensuring proper normalization, primary and foreign key relationships, and data integrity. The database served as a strong foundation for performing exploratory and advanced SQL queries, including revenue tracking, customer segmentation, and product performance analysis.

The analysis revealed important patterns in customer purchasing behavior, genre popularity, revenue concentration, and regional sales distribution. By identifying high-value customers, top-performing genres, and seasonal revenue trends, the project demonstrated how data-driven decision-making can enhance business strategy.

The interactive dashboard provided clear visualization of key performance indicators (KPIs) such as total revenue, total orders, total customers, monthly revenue trends, top customers, and most purchased genres. This enabled simplified communication of insights to non-technical stakeholders.

Overall, this project highlights the importance of combining database design, SQL analytics, and data visualization to support strategic planning and operational efficiency. The end-to-end analytical pipeline showcases practical skills relevant to real-world data analytics roles and demonstrates the ability to convert structured data into actionable business intelligence.